

The distribution of HPV vaccination coverage among adolescent girls in 6 European countries: *(The Netherlands, Belgium, Germany, France, Spain and Sweden)*

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Introduction

Cervical cancer is mainly caused by a viral infection: human papillomavirus (HPV). In 2019, the WHO published an estimate that HPV caused 620,000 cancer cases in women and 70,000 cancer cases in men. A persistent infection with high-risk HPV types (types 16 and 18) can lead to cancer. HPV vaccination is therefore an important preventative measure. The WHO recommends HPV vaccination for girls aged 9-14 as a key measure to prevent cervical cancer later in life. Despite the importance of early vaccination, in 2023 the global HPV vaccination coverage among girls receiving one dose was reported at 27 percent [3]. In this report, we focus on the research question: How did HPV vaccination coverage among adolescent girls change between 2010-2023 in 6 European countries (The Netherlands, Germany, Belgium, Spain, France and Sweden)?

Data sample

This question is answered, by using a public dataset provided by Our World in Data (which used WHO and UNICEF figures) which shows the share of girls in Europe who received their last HPV dose by age 15 [1] The dataset contains information on vaccination coverage across countries within the EU in years. Before conducting any analysis, Python (specifically the pandas package) was used to load the dataset and select the relevant country, year and vaccination coverage variables. The matplotlib package was the plotting library of choice. The data was also checked for missing values. The resulting dataset was then used to analyse the distribution of HPV vaccination in the 6 European countries, and 4 descriptive statistics were calculated (mean, standard deviation, skewness, excess kurtosis)

Results and discussion

The first plot (see Figure 1) displays the mean HPV vaccination coverage and its standard deviation. Mean coverage rose from about 26% in 2010 to about 68% in 2023, a substantial increase over the period. Most of the rise happened between 2011 and 2012, when the mean jumped to about 44%. While cleaning the data, I found that not all countries had reported data in 2010–2012, so the 2010–2011 means are not an accurate measure of coverage across all six countries. The light blue band shows the standard deviation, i.e. the variation in coverage between countries. It is narrow in 2010–2011, widens sharply in 2012, and then narrows again towards 2023. In 2023 the six countries sit about 16 percentage points above or below the mean, so coverage is far from identical, but the gap between countries has shrunk since 2012.

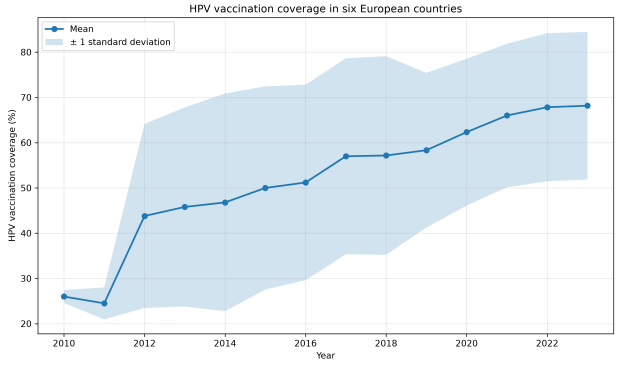


Figure 1: Mean HPV vaccination coverage (%) among girls (age 15) in six European countries, 2010–2023. The shaded band shows ±1 standard deviation.

The second plot (see Figure 2) shows the skewness and excess kurtosis of coverage across the six countries. The values for 2010–2011 (skewness 0, excess kurtosis -2) are not meaningful, because only a few countries had data. From 2012 to 2021 the skewness stayed between about -0.8 and -0.5, which means most countries had high coverage with a few lagging behind (a tail to the left). It then rose to about -0.3 in 2022–2023, suggesting a more symmetrical distribution. The excess kurtosis remained negative, between about -2.5 and -0.6, indicating a flatter distribution than a normal one. This suggests that coverage was fairly uniform across countries, with no extreme outliers. Though with solely analysing 6 countries, skewness and kurtosis are only rough indicators.

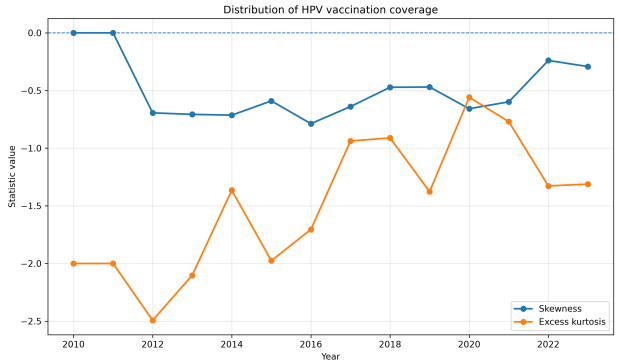


Figure 2: Skewness and excess kurtosis of HPV vaccination coverage across the six countries, 2010–2023.

Overall, HPV vaccination coverage among girls increased in the Netherlands, Germany, Belgium, Spain, France and Sweden between 2010 and 2023. It is important to note that I chose six high-income countries with well-funded health care systems in a similar region (Western and Northern Europe), which likely contributes to the similarity in coverage between them.

References

[1] Share of adolescent girls vaccinated against human papillomavirus. (2025). Our World in Data. `https://ourworldindata.org/grapher/coverage-of-the-human-papillomavirus-vaccine`

[2] WIISE Detail Page. (2025). Immunization Data. `https://immunizationdata.who.int/global/wiise-detail-page/human-papillomavirus-(hpv)-vaccination-coverage`

[3] World Health Organization. (2024, March 5). Human papillomavirus and cancer. `https://www.who.int/news-room/fact-sheets/detail/human-papilloma-virus-and-cancer`